HW2 Problem Set

ECS 240

Due: TBD

This problem set contains 5 problems. As with HW1, problems 1, 3, and 5 are coding problems, and problems 2 and 4 are pure theory problems. There is a template in `src/` which you can modify for the programming problems. Solutions to the theory problems should be written in LaTeX.

1.

2.

3.

4.

5.

Submission instructions:

- Upload your solutions (as a PDF) and your code (as a `.zip` file, as with HW0: including the entire repository directory including config files like `Cargo.toml`, but excluding `target/`) in Gradescope.
- Your PDF should be a single file `hw2.pdf`. If you use this LaTeX template to create your solutions, please remove the problem statements and include only your solutions.
- The `main.rs` should contain your solutions to all coding parts (including functions and types as described above). Please do not modify function signatures. You are welcome to add additional functions and tests – if you do so, please include all additional tests only at the bottom of the file.
- You may add utility modules, like `mod util;` then `use util::...`, if you feel that you need them; but you should put all the main functions with the names as above in `main.rs`. Please double check on Piazza before importing any external crates in your `Cargo.toml`.
- Make sure that `cargo build`, `cargo clippy`, `cargo test`, and `cargo fmt` (or `cargo fmt --check`) pass without any errors.